A Project Report on

GNN-BASED DYNAMIC USER PAIRING AND SLOT ASSIGNMENT FOR HYBRID NOMA NETWORKS

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

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by

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DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Name of Title: GNN-Based Dynamic User Pairing and Slot Assignment for Hybrid NOMA Networks

1. Abstract:

In modern wireless communication systems, the increasing number of users accessing the network simultaneously creates challenges in efficiently managing limited communication resources. Non-Orthogonal Multiple Access (NOMA) enables multiple users to share the same time and frequency resources, improving resource utilization and supporting higher network capacity. However, efficient user pairing and time-slot assignment remain challenging because of variations in channel conditions, interference, and user requirements. Improper pairing and resource allocation can reduce the achievable data rate, affect fairness, and increase the probability of communication failure. To address these challenges, this project presents a **Hybrid Graph Neural Network (GNN)-Based Approach for User Assignment and Slot Pairing in NOMA Networks**, combining graph-based learning, classical machine learning techniques, and wireless communication performance evaluation.

The proposed system is implemented in Python and uses simulated wireless network scenarios to evaluate user relationships and resource allocation strategies. Each user is represented as a node in a graph, with features such as user position, distance, channel gain, channel gain in decibels, signal-to-noise ratio (SNR), and minimum data-rate requirements. The GNN-based approach processes user relationships to support user pairing and slot assignment under the simulated network conditions. The system also incorporates Power-Domain NOMA (PD-NOMA) and Successive Interference Cancellation (SIC) for evaluating communication between paired users, along with a Sparse Code Multiple Access (SCMA) experiment to investigate its performance under different signal conditions. The hybrid scheduling approach is examined in relation to the implemented NOMA and SCMA components.

In addition to graph-based resource allocation, classical machine learning models, including Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regression, and Random Forest Regression, are evaluated for data-driven rate prediction. Their performance is compared using cross-validation and held-out test data, with metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). A paired bootstrap analysis is also used to estimate the uncertainty in the difference between prediction errors. These predictive modelling results are analyzed separately from the GNN's wireless communication performance to distinguish rate prediction accuracy from resource allocation effectiveness.

The wireless performance evaluation compares the GNN-based approach with conventional and reference methods, including Orthogonal Multiple Access (OMA), random pairing, near-far pairing, greedy allocation, and an optimization-based reference. Performance is assessed using metrics such as sum rate, average user rate, fairness, and outage probability. The experiments use simulated data with separate training, validation, and test scenarios for predictive modelling, allowing the implemented methods to be evaluated systematically. The results demonstrate the trade-offs between the compared approaches and help identify the strengths and limitations of the proposed system under the tested conditions. Overall, this project demonstrates the application of Graph Neural Networks and classical machine learning to user pairing, slot

assignment, and wireless resource allocation in simulated hybrid NOMA networks, while providing a comparative evaluation of prediction accuracy and communication performance.

2. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11
Programming Language: Python 3.10/3.11
Development Environment: Google Colab / Jupyter Notebook / VS Code
Machine Learning Framework: PyTorch
Graph Neural Network Library: PyTorch Geometric
Data Processing: NumPy, Pandas, Scikit-learn
Visualization: Matplotlib
Optional Interface: Streamlit

Signature of the Supervisor

Course Instructor